



Dehydrogenation of ethylbenzene to styrene over magnesium-doped hematite catalysts

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ABSTRACT

Styrene is an important building block of the plastics industry being produced by ethylbenzene dehydrogenation. The commercial catalysts have the disadvantages of deactivation by potassium loss and coke deposition, and chromium toxicity. To overcome these drawbacks, a new catalyst was developed by investigating the effect of magnesium on the catalytic properties of hematite in ethylbenzene dehydrogenation. Hematite was detected for all fresh catalysts and magnetite for the spent ones. Magnesium was found as magnesium ferrite and magnesium oxide, depending on the magnesium amount. Magnesium compounds affect the crystal sizes, the amount of defects and the reducibility of iron oxides, the predominance of each one depending on the composition of the solids. These effects caused important differences in activity, selectivity and stability of the catalysts. The catalyst with Mg/Fe = 0.09, consisting of aggregates of hematite, MgO and magnesium ferrite, showed the highest yield, being the most promising for commercial applications.

1. Introduction

Styrene is one of the most important monomers, being a basic building block of the plastics industry worldwide. It is used in the manufacture of plastics, resins, and elastomers, including polystyrene and acrylonitrile-butadiene styrene (ABS). According to Styrene Global Market Report 2022, the global styrene market is expected to grow from \$49.3 billion in 2021 to \$52.28 billion in 2022 and to \$62.81 billion in 2026. Most of the production of styrene is driven to polystyrene production. Polystyrene is used to manufacture ovens, microwaves, vacuum cleaners, and in the packaging industry. In the automobile industry, it is used in instrument panels, trim, energy-absorbing door panels, and sound-dampening foam [1].

Currently, most of the commercial production of styrene comes from the catalytic dehydrogenation of ethylbenzene, which accounts for around 90% of this monomer in the world [2]. The process suffers from

high energy consumption, equilibrium-limited ethylbenzene conversion, and catalyst deactivation [2,3]. This well-established technology is more than eight decades old, having received much attention and improvements over the years [4–6]. Several non-conventional catalysts have been proposed, such as tin-decorated ceria catalyst [7], nanodiamond-based monolithic foam catalyst [8] and three dimensional SiC@C hybrid catalyst [9], among others. Other studies have focused on the use of different oxidants, like carbon dioxide [2,4,5,10–16] or nitrous oxide [9,17], instead of steam. The use of a circulating fluidized bed reactor has also been proposed [18] to decrease the steam-to-ethylbenzene ratio. However, these processes are still under research as there are some technological barriers to their commercialization.

Under reaction conditions, other reactions occur simultaneously with ethylbenzene dehydrogenation (Eqs. 1 to 6) [3,5,19]. The main reaction (Eq. 1) is

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